



**Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 10 • STANDARD 6.GR.4

# Surface Area of Prisms

Lesson 10-4 • Teacher Copy

**Name:** \_\_\_\_\_

**Date:** \_\_\_\_\_

**Class:** \_\_\_\_\_

## My Notes

Level 3 Enrichment



### TODAY'S OBJECTIVES

**Content Objective:** I can find the surface area of rectangular and triangular prisms.

**Language Objective:** I can explain my work using the words surface area, rectangular prism, triangular prism, and face.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



### WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Surface area

Rectangular prism

Triangular prism

Face

Net

Lateral face



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

The total area of all the faces of a solid is its .

2

A solid with six rectangular faces is a .

3

A solid with two triangular faces and three rectangular faces is a

.

4

A flat surface of a solid figure is a .

5

A flat pattern that folds into a solid is a .

6

A side face of a prism that is not a base is a .

## Write About the Math

### Guided ESOL writing

Use the support level you need. Every level answers the same math question.

### 1. Understand the Question

#### EXPLAIN

**For the triangular prism (base 6, height 4, length 10, slant sides 5), how did you account for the triangular bases AND the rectangular lateral faces?**

**Your job:** Answer the question. Use math evidence. Explain how the evidence proves your answer.

*Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.*

### 2. Plan Your Math Words

Check at least two words you will use.

☐

**Surface area** — The total area of all the flat sides of a solid.

*Área de superficie — El área total de todos los lados planos de un sólido.*

☐

**Triangular prism** — A solid with two triangle ends and three flat rectangle sides.

*Prisma triangular — Un sólido con dos extremos triangulares y tres lados rectangulares.*

☐

**Face** — One flat side of a solid shape.

*Cara — Un lado plano de una figura sólida.*

☐

**Net** — A flat shape that folds up into a solid.

*Plantilla (desarrollo plano) — Una figura plana que se dobla y forma un sólido.*

☐

**Lateral face** — The side faces of a solid, not the top or bottom.

*Cara lateral — Los lados de un sólido, no la parte de arriba ni la de abajo.*

**Say it first:** Say your idea to a partner or quietly to yourself before you write.

*Di tu idea a un compañero o en voz baja antes de escribir.*

**The two triangular bases together are  $2 \times (\frac{1}{2} \times 6 \times 4) = \underline{\hspace{2cm}}$ .**

*Las dos bases triangulares juntas son  $2 \times (\frac{1}{2} \times 6 \times 4) = \underline{\hspace{2cm}}$ .*

### 3. Build Your Explanation

**Model the parts:** This model shows the parts of an explanation. It does not answer your writing question.

**Claim:** For both, you add the areas of all the faces, but the shapes and number of faces differ.

**Evidence:** Students explain both methods add up all face areas, but a rectangular prism has 6 rectangular faces while a triangular prism has 5 faces (2 triangles + 3 rectangles).

**Reasoning:** This evidence connects the definition of surface area to the mathematical claim.

#### Start

Most language support

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- **The two triangular bases together are**  $2 \times (\frac{1}{2} \times 6 \times 4) = \underline{\hspace{2cm}}$ .  
*Las dos bases triangulares juntas son*  $2 \times (\frac{1}{2} \times 6 \times 4) = \underline{\hspace{2cm}}$ .
- **The three lateral faces add to**  $\underline{\hspace{2cm}}$ , **so the total surface area is**  $\underline{\hspace{2cm}}$ .  
*Las tres caras laterales suman*  $\underline{\hspace{2cm}}$ , *así que el área total de superficie es*  $\underline{\hspace{2cm}}$ .

#### Build

Some language support

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- **My claim is**  $\underline{\hspace{2cm}}$ .  
*Mi afirmación es*  $\underline{\hspace{2cm}}$ .
- **I know because**  $\underline{\hspace{2cm}}$ .  
*Lo sé porque*  $\underline{\hspace{2cm}}$ .
- **So**  $\underline{\hspace{2cm}}$ .  
*Entonces*  $\underline{\hspace{2cm}}$ .

## Explain

### Light language support

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

- **My claim is \_\_\_\_.**  
*Mi afirmación es \_\_\_\_.*
- **My math evidence is \_\_\_\_.**  
*Mi evidencia matemática es \_\_\_\_.*
- **This proves \_\_\_\_ because \_\_\_\_.**  
*Esto demuestra \_\_\_\_ porque \_\_\_\_.*

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## 4. Check Your Explanation

- ☐ I answered the question.  
*Respondí la pregunta.*

- ☐ I used math evidence: numbers, an equation, a model, or a comparison.  
*Usé evidencia matemática: números, una ecuación, un modelo o una comparación.*

- ☐ I explained how the evidence proves my answer.  
*Explicué cómo la evidencia demuestra mi respuesta.*

- ☐ I used at least two math words.  
*Usé por lo menos dos palabras matemáticas.*

- ☐ I reread complete sentences and punctuation.  
*Volví a leer mis oraciones completas y la puntuación.*

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## Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

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## Answer Key & Teacher Guide

1. **Notes 1:** Surface area
2. **Notes 2:** Rectangular prism
3. **Notes 3:** Triangular prism
4. **Notes 4:** Face
5. **Notes 5:** Net
6. **Notes 6:** Lateral face
7. **Watch for this mistake:** *A common mistake in Surface Area of Prisms is confusing surface area with volume — multiplying  $l \times w \times h$  (which finds the space INSIDE the prism, in cubic units) instead of adding the areas of all the faces ( $SA = 2lw + 2lh + 2wh$ , in square units). A second common slip is forgetting to double one face pair, like writing  $2(l \times w) + 2(l \times h) + (w \times h)$  instead of  $+ 2(w \times h)$  — always count all 6 faces (or all 5 for a triangular prism) before you submit.*
8. **Try It 1:** A.  $148 \text{ cm}^2$  —  $SA = 2(6 \times 5) + 2(6 \times 4) + 2(5 \times 4) = 60 + 48 + 40 = 148 \text{ cm}^2$ .
9. **Try It 2:** A.  $288 \text{ in}^2$  — *Two triangle ends:*  $2 \times (\frac{1}{2} \times 8 \times 6) = 48 \text{ in}^2$ . *Three rectangles:*  $(8 + 6 + 10) \times 10 = 24 \times 10 = 240 \text{ in}^2$ .  $SA = 48 + 240 = 288 \text{ in}^2$ .